

HTML CSS JS Test Questions

1. What is HTML and why is it important?
 2. What are tags in HTML?
 3. What is the difference between `<div>` and `<span>`?
 4. What are semantic HTML elements?
 5. Explain the structure of a basic HTML document.
 6. How do you embed multimedia in HTML?
 7. What is the role of attributes in HTML tags? Give examples.
 8. What is the purpose of the `<head>` tag?
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1. What is CSS?
 2. What is the difference between classes and IDs in CSS?
 3. What is the Box Model in CSS?
 4. What is z-index in CSS?
 5. How can you center a div?
 6. How can you center a div horizontally and vertically?
 7. What's the difference between `min-width`, `max-width`, and `width`?
 8. What are media queries? How do you make a layout responsive using media queries?
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1. What is Javascript why it's need web development?
 2. What is mean data and type of datatypes? List out it's types.
 3. What is the difference between `var`, `let`, and `const`?
 4. What is hoisting in JavaScript?
 5. What is the difference between `==` and `===`?
 6. What are the operations can you do in string?
 7. What does an array mean? Write one array example. And list out the operations
 8. What are the operations can you do in string?
 9. What is a callback function?
 10. Write scheduling function js.
 11. Write fetching method in js

General Instructions:

Attempt all questions unless instructed otherwise.

Write your name, batch, course clearly on the top of the answer sheet.

I need your full attention, and your answers must be clear, neat, and written in your own understanding using the terms and techniques taught in class.

Do not talk to other students or look at anyone else's paper during the test.

If you have any doubts, mark the question and we will discuss it after the test.

Web Development Interview Q&A Document

HTML (HyperText Markup Language)

Q1: What is HTML and why is it important?

HTML is the standard markup language used to create web pages. It provides structure to web content and is essential for rendering content in browsers.

Q2: What are tags in HTML?

Tags are keywords enclosed in angle brackets (e.g., <p>) that define elements and instruct the browser how to display content.

Q3: What is the difference between <div> and ?

- **<div>: Block-level element for grouping larger sections.**
- **: Inline element used for small chunks of text.**

Q4: What are semantic HTML elements?

Semantic elements clearly describe their meaning, improving readability and accessibility (e.g., <header>, <footer>, <nav>, <article>).

Q5: Explain the structure of a basic HTML document.

<!DOCTYPE html>

<html>

```
<head>

  <title>Title</title>

</head>

<body>

  <h1>Hello, world!</h1>

</body>

</html>
```

Q6: How do you embed multimedia in HTML?

- Image: ``
- Audio: `<audio controls src="audio.mp3"></audio>`
- Video: `<video controls src="video.mp4"></video>`

Q7: What is the role of attributes in HTML tags? Give examples.

Attributes provide additional information. Example: `<a href="https://example.com">Link</a>`

Q8: What is the purpose of the `<head>` tag?

It holds metadata, links to CSS/JS files, page title, and character encoding.

CSS (Cascading Style Sheets)

Q9: What is CSS?

CSS styles HTML elements including layout, colors, and fonts.

Q10: What is the difference between classes and IDs in CSS?

- Class (.): Reusable, e.g., `.button`
- ID (#): Unique, used once per page, e.g., `#header`

Q11: What is the Box Model in CSS?

It includes content, padding, border, and margin, defining how elements occupy space.

Q12: What is z-index in CSS?

It determines the stack order of elements. Higher z-index means the element appears on top.

Q13: How can you center a div?

Horizontally: `margin: 0 auto;`

Using Flexbox:

`display: flex;`

`justify-content: center;`

Q14: How can you center a div horizontally and vertically?

`display: flex;`

`justify-content: center;`

`align-items: center;`

`height: 100vh;`

Q15: What's the difference between min-width, max-width, and width?

- **width:** Fixed width
- **min-width:** Minimum size
- **max-width:** Maximum size

Q16: What are media queries?

They apply styles based on screen size. Example:

`@media (max-width: 768px) {`

`body {`

`background: lightblue;`

```
}  
  
}
```

JavaScript (JS)

Q17: What is JavaScript and why is it needed in web development?

JavaScript adds interactivity to websites (e.g., form validation, animations, user events).

Q18: What is data and what are the types of data types in JS?

Data represents values. Types:

- **Primitive:** String, Number, Boolean, Null, Undefined, Symbol, BigInt
- **Non-Primitive:** Object, Array, Function

Q19: What is the difference between var, let, and const?

- **var:** Function-scoped, hoisted
- **let:** Block-scoped, can be updated
- **const:** Block-scoped, cannot be reassigned

Q20: What is hoisting in JavaScript?

Variables and functions are moved to the top of their scope before execution.

Q21: What is the difference between == and ===?

- **==:** Loose equality (type conversion)
- **===:** Strict equality (no conversion)

Q22: What operations can you do in strings?

length, charAt(), toUpperCase(), toLowerCase(), substring(), replace(), split(), trim()

Q23: What does an array mean? Write one array example and list operations.

Array: Ordered collection.

Example: `let fruits = ["apple", "banana", "mango"]`

Operations: `push()`, `pop()`, `shift()`, `unshift()`, `map()`, `filter()`, `sort()`

Q24: What is a callback function?

A function passed into another function to run after it completes.

```
function greet(name, callback) {  
  
    console.log("Hi " + name);  
  
    callback();  
  
}  
  
function sayBye() {  
  
    console.log("Bye!");  
  
}  
  
greet("Ganapathi", sayBye);
```

Basic HTML Interview Questions

1. What is HTML?
2. What are HTML tags?
3. What is the difference between HTML and XHTML?
4. What are semantic elements in HTML?
5. What is the difference between `<div>` and `<span>`?
6. What are empty (void) elements in HTML?
7. What is the purpose of the `<!DOCTYPE>` declaration?
8. What is the difference between block-level and inline elements?
9. What are attributes in HTML?
10. What is the difference between `id` and `class`?

HTML5 Related Questions

1. What is new in HTML5?
2. What are semantic tags introduced in HTML5?
3. What is the difference between `<section>` and `<article>`?
4. What is the `<canvas>` element?
5. What is the `<audio>` and `<video>` tag?
6. What are data attributes in HTML5?
7. What is localStorage and sessionStorage?
8. What is the difference between `<figure>` and `<img>`?
9. What is the `<datalist>` tag?

10. What are input types introduced in HTML5?

Forms & Input Questions

1. What are the different types of input elements?
 2. What is the difference between **GET** and **POST**?
 3. What is form validation?
 4. What are required, pattern, and placeholder attributes?
 5. What is the difference between **readonly** and **disabled**?
 6. What is the purpose of the **<label>** tag?
 7. How do you upload a file using HTML form?
 8. What is **enctype** attribute in form?
 9. How does autocomplete work?
 10. What is the difference between checkbox and radio button?
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◆ Tables & Lists

1. What are different types of lists in HTML?
2. What is the difference between **** and ****?
3. What is **<dl>** tag?
4. How do you merge table cells?
5. What is the difference between **<thead>**, **<tbody>**, and **<tfoot>**?
6. What is **colspan** and **rowspan**?

◆ Advanced HTML Questions

1. What is the DOM?
2. What is the difference between HTML and DOM?
3. What are meta tags?
4. What is viewport in HTML?
5. What is the difference between `async` and `defer` in script tag?
6. What is SEO and how does HTML help in SEO?
7. What is accessibility in HTML?
8. What are ARIA attributes?
9. What is the difference between inline, internal, and external CSS?
10. How does browser rendering work?

🔥 Very Important Practical Questions

1. Create a basic HTML page structure.
2. Create a registration form.
3. Create a table with merged cells.
4. Create a navigation bar using semantic tags.
5. Embed a YouTube video in HTML.
6. Create a responsive image using `srcset`.
7. Create a dropdown using `<select>` and `<option>`.

◆ Basic CSS Interview Questions

1. What is CSS?
 2. What are the different types of CSS? (Inline, Internal, External)
 3. What is the CSS Box Model?
 4. What is the difference between margin and padding?
 5. What is the difference between `id` and `class` in CSS?
 6. What are CSS selectors?
 7. What is specificity in CSS?
 8. What is `!important` in CSS?
 9. What are pseudo-classes?
 10. What are pseudo-elements?
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◆ Selectors & Specificity Questions

1. What is the difference between `.class` and `#id` selector?
2. What is attribute selector in CSS?
3. What is the difference between child selector and descendant selector?
4. What is the universal selector?
5. What is grouping selector?

6. What is combinator in CSS?
 7. What is CSS specificity hierarchy?
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◆ Box Model & Layout Questions

1. Explain CSS Box Model with diagram.
 2. What is `box-sizing: border-box`?
 3. What is the difference between `display: none` and `visibility: hidden`?
 4. What is the difference between inline, block, and inline-block?
 5. What is position property in CSS?
 6. What is the difference between relative, absolute, fixed, and sticky?
 7. What is float in CSS?
 8. What is clear property?
 9. What is z-index?
 10. What is overflow property?
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◆ Flexbox Interview Questions

1. What is Flexbox?
2. What is the difference between `justify-content` and `align-items`?
3. What is flex container and flex item?

4. What is `flex-grow`, `flex-shrink`, `flex-basis`?
 5. What is the difference between `row` and `column` in flex-direction?
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◆ CSS Grid Questions

1. What is CSS Grid?
 2. Difference between Flexbox and Grid?
 3. What is `grid-template-columns`?
 4. What is `fr` unit?
 5. What is grid area?
 6. What is the difference between auto-fill and auto-fit?
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◆ Responsive Design Questions

1. What is media query?
 2. What is mobile-first approach?
 3. What is viewport?
 4. What is responsive web design?
 5. What is rem, em, %, vh, vw?
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◆ **Advanced CSS Questions**

1. What is CSS preprocessor? (SASS, LESS)
 2. What is CSS variable?
 3. What is animation in CSS?
 4. What is transition property?
 5. What is transform property?
 6. What is keyframes?
 7. What is shadow DOM?
 8. What is BEM methodology?
 9. What is critical CSS?
 10. What are CSS frameworks?
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Practical Coding Questions (Very Important)

1. Create a responsive navbar using Flexbox.
2. Center a div horizontally and vertically.
3. Create a card layout using Grid.
4. Create a simple animation using keyframes.
5. Make an image responsive.
6. Create a sticky header.
7. Create a dropdown menu using only CSS.

React Js Test Questions

1. What is React? How is it different from traditional JavaScript frameworks?
2. What is the virtual DOM and how does React use it?
3. What are the main features of React?
4. What is JSX? Can a browser read JSX directly?
5. What are components in React? Write basics syntax
6. What is the purpose of key in lists?
7. How do you pass data from parent to child in React?
8. What is the difference between **state** and **props**?
9. What is a Hook in React?
10. What are the main lifecycle methods in a class component?

11. List out the Major and main concepts in Reactjs(Key Words)

1. Introduction & Fundamentals

1. What is React.js?
2. What are the main features of React?
3. What is the difference between a class component and a functional component?
4. What is JSX and why is it used?
5. What are components in React?
6. What is the difference between state and props?
7. What is virtual DOM and how does it work?

8. What is the purpose of key in React lists?
 9. Explain how React updates the UI efficiently.
 10. What are React Fragments and why are they used?
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2. React Hooks

1. What are hooks in React?
 2. Explain `useState()` with an example.
 3. Explain `useEffect()` with an example.
 4. What is the difference between `useEffect()` and lifecycle methods?
 5. What is `useRef()` used for?
 6. What is `useMemo()` and when to use it?
 7. What is `useCallback()` and how is it different from `useMemo()`?
 8. How to create a custom hook?
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3. Props, State & Lifecycle

1. How do you pass data from parent to child components?
2. How can you pass data from child to parent components?
3. What are controlled and uncontrolled components?
4. Explain component lifecycle methods in class components (`componentDidMount`, `componentDidUpdate`, etc.).
5. What is lifting state up in React?

4. Rendering & Events

- 1. What is conditional rendering in React?**
 - 2. What is list rendering?**
 - 3. How to handle forms in React?**
 - 4. How do you handle events in React?**
 - 5. What are synthetic events in React?**
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5. Routing

- 1. What is React Router?**
- 2. How to implement routing in React?**
- 3. What is <Link> vs <a> tag?**
- 4. What is a route parameter?**
- 5. What is `useNavigate()` in React Router v6?**

7. API Handling

- 1. How do you fetch data from an API in React?**
 - 2. How do you handle loading and error states during an API call?**
 - 3. What is Axios and how is it used in React?**
 - 4. How do you display data fetched from JSON?**
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8. Performance & Optimization

- 1. How to optimize performance in React apps?**
- 2. What is React.memo()?**
- 3. What are higher-order components (HOC)?**
- 4. What are pure components?**
- 5. What is lazy loading and how is it implemented?**